



Heat Transfer in Liquid Metal

Brian Bertacco, Antonio Morrone, Giulio Riva
May 12, 2026 — KTH Royal Institute of Technology



Outline

Introduction

Single Rod Heater

Turbulence Modelling

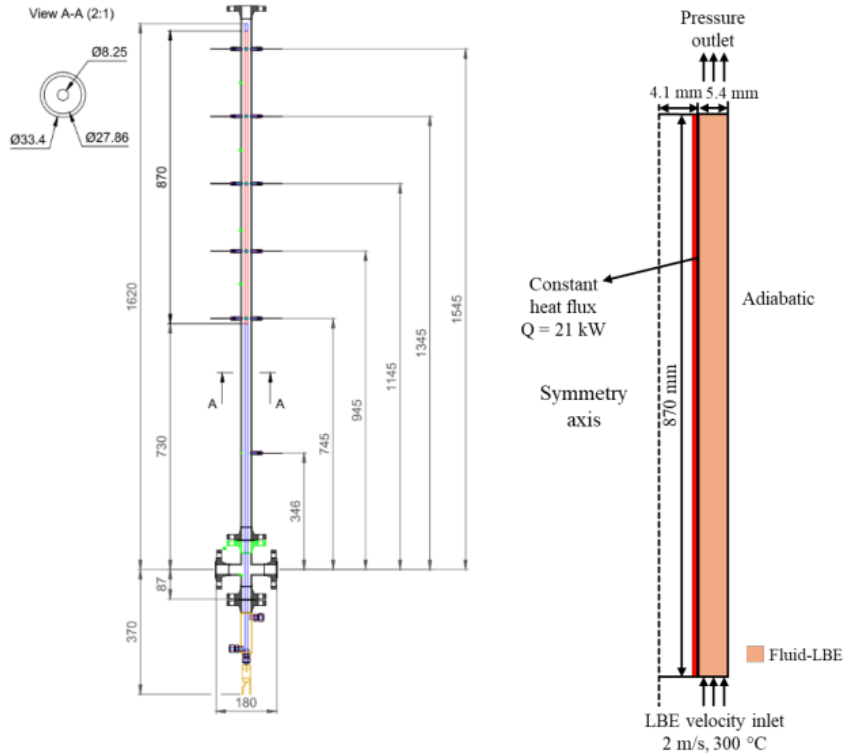
Straight Tube Heat Exchanger

Conclusions

The Liquid Metal Modelling Challenge

- **Lead-Bismuth Eutectic (LBE):** A prominent coolant candidate for Gen IV Fast Reactors, characterized by high thermal conductivity and an extremely low Prandtl number ($Pr \approx 0.02 - 0.03$).
- **Conduction Dominance:** Unlike conventional fluids, molecular heat conduction in LBE remains the dominant heat-transfer mechanism even under highly turbulent flow conditions.
- **Boundary Layer Separation:** Classical forced-convection theory dictates that $\frac{\delta_v}{\delta_T} \sim Pr^{1/3}$. For $Pr \ll 1$, the thermal boundary layer becomes significantly thicker than the viscous one.
- **The RANS Limitation:** Standard turbulence models are calibrated for $Pr \approx 1$ fluids (e.g., water, air). Consequently, they struggle with the separated boundary layers in LBE and tend to severely over-predict the turbulent heat flux.

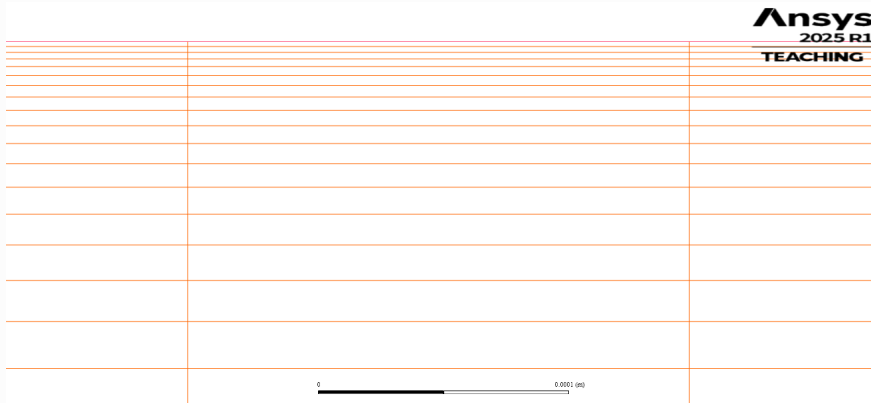
Geometry & Setup



Model Specifications:

- 2-D axisymmetric annular channel.
- **Active Length:** 870 mm with 21 kW total heat power.
- **Inlet:** 100 mm unheated section for flow development.
- Temperature-dependent properties implemented via UDFs.

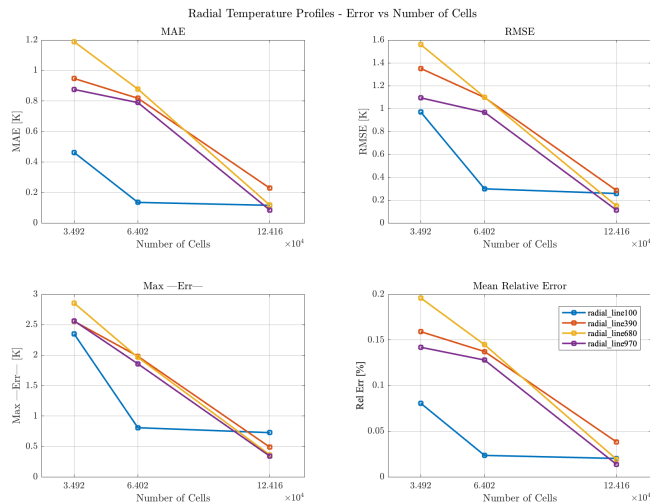
Mesh Generation Strategy



Discretization Approach:

- **Refinement Levels:** Four distinct grids generated, ranging from coarse to highly refined.
- **Target y^+ :** Explicit resolution of the viscous sublayer ($y^+ \approx 1$) is strictly required for the low-Re $k - \omega$ SST formulation.
- **Near-Wall Sizing:** Achieved the target y^+ with a calculated first-cell height of $\Delta y_1 \approx 2 \mu\text{m}$.

Grid Convergence & Error Analysis



Statistical Evaluation:

- Methodology:** Temperatures from coarser meshes were linearly interpolated onto the 388k-cell grid coordinates to directly compare errors.
- Stability:** Mean Relative Error (MRE) exhibited clear convergence and remained extremely low ($< 0.2\%$) across all grids.
- Optimal Selection:** Mesh 2 (124k cells) was selected as it successfully reduced the max absolute error to < 1 K while balancing cost.

Turbulent Prandtl Number (Pr_t)

Two distinct modelling approaches evaluated:

Reynolds Modification

(Global Approach)

- Single value computed from bulk Re and Pe .
- Enforces a constant $Pr_t \approx 1.8$.
- **Impact:** Artificially suppresses radial thermal mixing across the entire domain.

Kays Modification

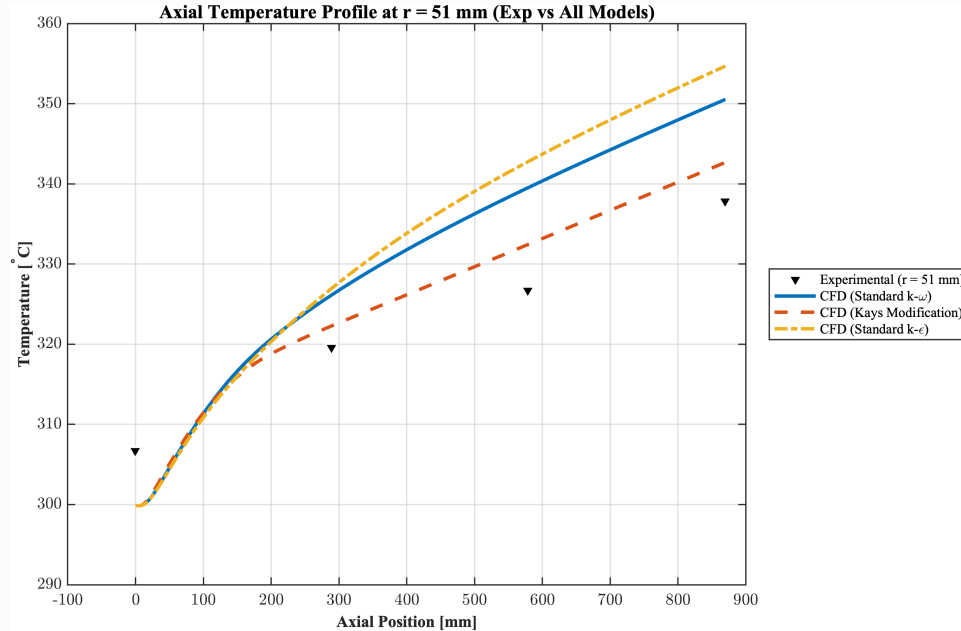
(Dynamic/Local Approach)

- Dynamically resolved based on local turbulent Péclet number (Pe_t):

$$Pr_t = 0.85 + \frac{0.7}{Pe_t}$$

- **Impact:** Realistically enhances effective thermal diffusivity in the boundary layer transition region.

Single Rod Validation



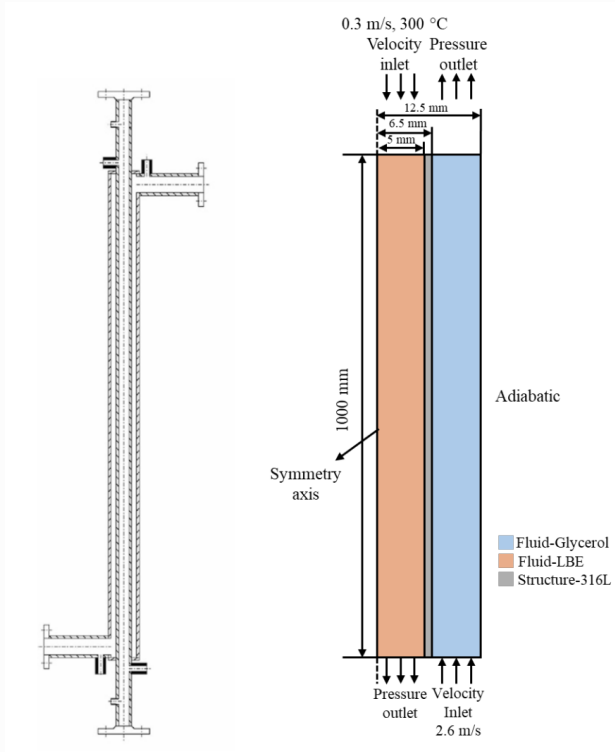
Model Performance:

- **Standard $k - \epsilon$:** Severe temperature over-prediction.
- **$k - \omega$ SST:** Slight improvement due to sublayer resolution, but still overestimates heat accumulation.
- **Kays Modification:** Best overall fit. Closely traces experimental data by successfully mitigating artificial wall heat buildup.

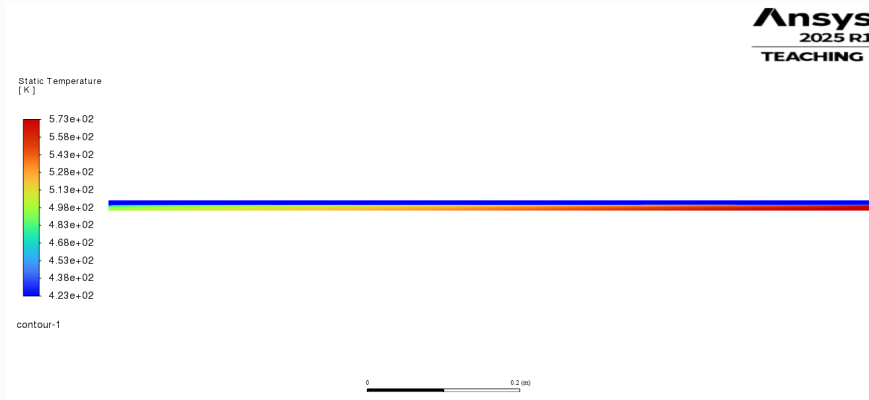
Conjugate Heat Transfer Setup

Model Specifications:

- 3 coupled domains: LBE, 316L SS wall, and Glycerol.
- Counter-current arrangement with conformal mesh.
- **LBE:** Inlet at 300°C.
- **Glycerol:** Inlet at 150°C (Transitional/turbulent flow regime).



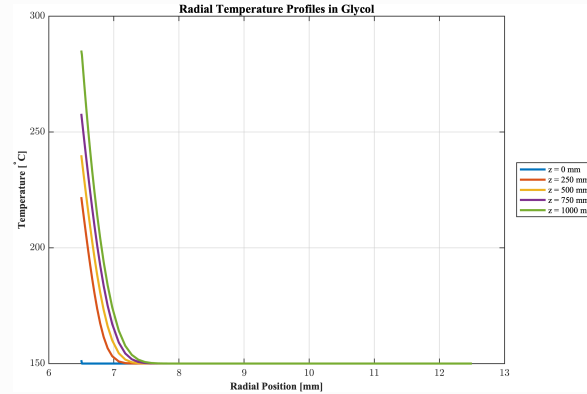
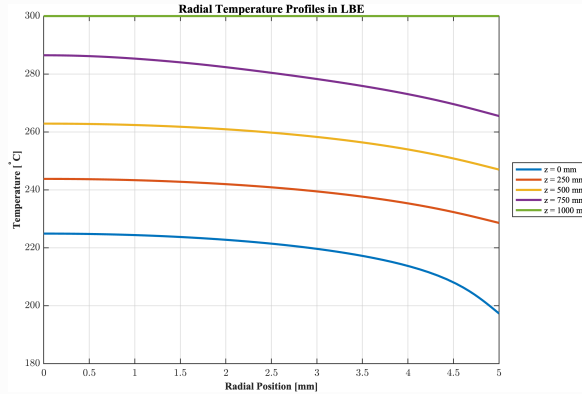
Heat Exchanger Results



The Glycerol Thermal Sink:

- Glycerol heat capacity rate $\gg$ LBE heat capacity rate.
- Glycerol bulk temperature remains flat; it acts as an **infinite thermal sink**.
- High Pr (≈ 45) of glycerol confines radial heat exchange to an extremely thin boundary layer.

Radial Heat Transport: LBE vs. Glycerol



Observations:

- **LBE (Left):** Low- Pr ensures a gentle temperature gradient across the bulk.
- **Glycerol (Right):** High- Pr fluid results in extremely severe radial gradients.
- **Thermal Sink:** Heat is strictly confined to a thin boundary layer, leaving the bulk unmixed.

Sources of Discrepancy

Key physical and numerical uncertainties:

- **Unmodelled Heat Losses:** The adiabatic assumption ignores real-world parasitic thermal losses to the ambient, leading to over-predicted LBE temperatures.
- **Transient vs. Steady-State:** Experimental data reflects a cold-start transient ($t = 2000$ s) where the bulk LBE still absorbs sensible heat, mismatching the assumed instantaneous steady state.
- **Geometric & Property Simplifications:** 2-D axisymmetric models neglect 3-D mixing vortices, while LBE thermal conductivity correlation carries an inherent $\pm 15\%$ experimental uncertainty.

Final Conclusions & Future Work

- **Standard RANS Limitations:** Standard models fundamentally fail to capture the highly decoupled momentum and thermal mixing of liquid metals.
- **Dynamic Pr_t (Kays):** Calculating a local Pr_t successfully enhances effective thermal diffusivity in the boundary layer, mitigating the artificial wall heat accumulation.
- **The Thermal Bottleneck:** Despite acting as an infinite thermal sink, the glycerol's high Pr (≈ 45) confines heat exchange to an extremely thin layer, acting as the primary thermal resistance.
- **Future Advancements:** Strict validation requires replacing standard models with ad hoc liquid metal turbulence formulations and implementing realistic, non-adiabatic boundary conditions.